

COUNSELING IN FETAL MEDICINE: AGENESIS OF THE CORPUS CALLOSUM

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Abstract: The aim of this review is to provide up-to-date and evidence based answers to common questions regarding the diagnosis and prognosis of prenatally detected absence of the corpus callosum (ACC). A systematic literature search was performed to identify all reports of ACC and reference lists of articles were identified. ACC involves partial or complete absence of the main commissural pathway that connects the two cerebral hemispheres, and can be isolated (with no other abnormalities) or complex (with other abnormalities coexisting). It is a rare finding and the prevalence is difficult to estimate because of selection bias in reported series. The corpus callosum (CC) can be assessed on ultrasound by direct visualisation, but indirect features such as ventriculomegaly, absence of the cavumseptipellucidi or widening of inter-hemispheric fissure are commonly the reason for detection in a screening population. Careful imaging in a centre with a high level of expertise is required to make a full assessment and to exclude coexisting abnormalities, which may occur in about 46% of fetuses. Where available, MRI appears to be an important adjunct as it allows direct visualization. It can reduce false positive rates on ultrasound and can confirm ACC, it can assess whether this is complete or partial, and it can help in detecting coexisting brain abnormalities not seen on ultrasound. The overall rate of chromosomal abnormality in fetuses with ACC is 18%, but this high rate includes both isolated and complex ACC; more recent studies suggest that chromosomal abnormalities are rare in isolated cases. Nevertheless, postnatal follow up studies suggest that about 15% of cases thought to be isolated prenatally were found to have associated abnormalities after birth. Neurodevelopmental outcome in isolated ACC was recently reported in a systematic review and suggested normal outcome in about 65-75% of cases. The findings need to be seen in light of the several limitations of existing studies, in terms of study design, selection bias, varying definitions and imaging protocols, ascertainment bias and lack of control groups. These uncertainties mean that antenatal counselling remains difficult and further large prospective studies are needed.

INTRODUCTION

Agenesis of the corpus callosum (ACC) is a rare condition where the main commissural pathway that connects the two cerebral hemispheres is partially or completely absent. It can be detected on antenatal ultrasound, and poses a significant diagnostic challenge as the outcome is variable¹. This depends on whether there are coexisting abnormalities and the underlying cause, but even in isolated ACC outcomes vary. In this review we aim to answer some common questions we encounter in our daily practice using the available literature.

METHODS

A list of common questions we encounter in our practice was assembled from fetal medicine consultants working in the unit. A systematic search strategy was then employed in order to identify all those articles that gave relevant information. A Pubmedsearch was conducted on the 5/05/2011 and updated on the 30/07/2012. Over 3600 titles and abstracts were reviewed. The search strategy, keywords used and results are found in Appendix 1.

1. WHAT IS THE CORPUS CALLOSUM?

The corpus callosum (CC) is the main commissural pathway in the brain and has an important role in the integration of information between the two hemispheres. It has four segments: the rostrum, genu, body and splenium; the narrowing between the body and splenium is called the isthmus (Fig 1).

The formation of the CC starts with the development of the genu; the body, the isthmus and the splenium develop at a later stage². The development of the CC is thought to progress cranio-caudally with the exception of the most anterior part called the rostrum that develops later, although other studies suggest that the development of the CC starts with the formation of the anterior body and progresses bidirectionally³⁻⁸.

By 18-20 weeks of gestation, the CC assumes its final shape, although some thickening continues^{9,10}. As the CC does not assume its final shape until 20 weeks, imaging of this structure at earlier gestations may not be optimal.

This developmental sequence is thought to be important as it may point to the aetiology: if partial ACC involves the anterior portion it is usually due to a disruptive event (such as due to vascular events or infectious causes)^{11,12}. Conversely, if the posterior portion is absent it may suggest an arrest in the development¹¹.

2. WHAT ARE THE POSSIBLE ABNORMALITIES OF THE CC?

There is wide variation in the terminology used to describe CC abnormalities, including: hypoplasia (thinning of the CC), hyperplasia (thickening of the CC) and agenesis (absence of the CC). When agenesis occurs this can be complete (absence of all components) or partial (presence of a short remnant). Dysgenesis and hypogenesis (incomplete development) also refer to partial ACC^{2,13-15}. Such differences in terminology should be considered when assessing reports of outcomes in this condition: although in many papers abnormalities of the CC are grouped, hypoplasia, complete agenesis and partial agenesis may be different clinical entities. Hypoplasia is rarely diagnosed antenatally, and most of the available series focus upon agenesis, partial or complete.

In addition to the above, major malformations of the forebrain can prevent the formation of the corpus callosum, such as encephaloceles and holoprosencephaly; in these situations the outcome is of course dictated not by the ACC itself but by the primary abnormality.

TERMINOLOGY USED IN THIS PAPER

- Partial agenesis – Partial absence of the CC
- Complete agenesis – Total absence of the CC

Each of these may be

- Isolated – ACC with no other abnormalities
- Complex – ACC with other abnormalities

3. HOW COMMON IS ACC?

The incidence is difficult to estimate because of selection bias in reported series. Large studies in healthy individuals using magnetic resonance imaging (MRI) showed no cases of ACC¹⁶⁻¹⁹. Similarly, two postmortem series involving 59000 autopsies

found ACC in only three cases^{20,21}. However, it is important to note that assessment in healthy adults, study volunteers or routine autopsies is not unselected as these populations may be less likely to suffer from ACC. The best available data probably come from the California Birth Defect Monitoring Program and suggest a prevalence of about 1.4 and 0.4 per 10,000 live births for ACC and hypoplasia of CC respectively²². This figure may, however, be an underestimate of the real incidence as it is likely that a proportion of asymptomatic individuals with ACC escaped detection in this study. These data suggest that anomalies of the CC may be quite frequent congenital anomalies of the central nervous system, with an overall incidence similar to neural tube defects. A higher prevalence of ACC in males has been reported by some but not other studies²²⁻²⁵.

4. HOW CAN WE RELIABLY IMAGE THE CORPUS CALLOSUM ON ULTRASOUND?

Normal ultrasound appearance of the CC

In median views of the fetal brain (2D ultrasound) the CC appears as a thin anechoic space, lined superiorly and inferiorly by two echogenic lines (Fig.1)¹³. Sonographic demonstration of the CC requires an adequate angle of insonation; median and mid coronal scans of fetal brain are the best planes to visualize CC and can be obtained with standard transabdominal ultrasound if the fetus is breech or with transvaginal ultrasound if the fetus is in vertex presentation^{13,26,27}. Multiplanar sonography and transvaginal ultrasound may be useful in establishing the presence of the CC^{28,29}. The pericallosal artery, which develops in close association with the CC, can also act as a useful marker on 2D ultrasound imaging (Fig1); this may be particularly useful when encountering resolution-related difficulties due to, for example, maternal obesity or fetal position. Offline analysis of three-dimensional (3D) volumes may also allow visualisation, but is dependent on the plane of volume acquisition. 4D volume contrast from the C-plane (VCI-C) is also effective in real time reconstruction of median planes from axial views³⁰. Comparison of 2D with 3D and 4D VCI-C suggests that the quality of the 2D is superior, and that 3D ultrasound, while easy to perform, does not overcome all 2D limitations³⁰.

ACC: direct visualisation

The diagnosis of ACC is based upon a non-visualization of the CC. If underdeveloped the measurement of its length and thickness is suggested. Normative charts for comparison are available^{10,28}. However, there are no data indicating which is the best threshold to diagnose partial agenesis or abnormal thickness. The available normograms report the 90 % prediction interval, and it would not be appropriate to use the threshold of the 5th centile to diagnose a rare condition. In practice, it is easy to recognize abnormal size of the CC only when there is a gross alteration. This means that cases with slight deviations from normal values, particularly in early gestation, represent a diagnostic dilemma.

ACC: indirect features

In practice it can be difficult to positively identify presence or absence of the CC. However, ACC causes indirect brain rearrangements and these are often the first clues to the diagnosis (see box 1). These indirect signs are not always encountered and are inconsistently found in fetuses with partial agenesis of the CC^{2,31}. Ghiat *al* showed that absence of the CSP or ventriculomegaly were detected in 8 out of 14 cases of partial agenesis of the CC; other signs of ACC were not demonstrated and this may reflect a minor disruption of the midline fibres³¹.

5: HOW GOOD IS ANTENATAL ULTRASOUND IN THE DETECTION OF ACC?

Prenatal sonographic diagnosis of agenesis of the CC is difficult. In expert hands, when a neurosonographic examination is performed, complete ACC may be diagnosed in the majority of cases at midgestation²⁷. Nevertheless, in the standard examination of low risk fetuses, visualization of the CC is not required and the condition may be suspected only in the presence of indirect findings, including ventriculomegaly or failure to visualise the cavum septi pellucidi³²⁻³⁵. Detection may also be gestational age dependent, and in one series the diagnosis could never be made in early gestation³⁶.

6. DISTINGUISHING ISOLATED FROM COMPLEX CASES

ACC has been associated with cranial and extra cranial anomalies, and there is an association with chromosomal and genetic abnormalities. Some of these associated conditions are subtle and difficult to recognize antenatally and careful imaging in a

centre with a high level of expertise is required. In our review we found a prevalence of associated brain abnormalities of 45.8% (Appendix2). It is important to note that this rate is likely to be higher than the true figure due to bias: truly isolated cases may escape detection while those with associated abnormalities are detected, referred to specialist centres and reported in the literature. The most frequently reported brain abnormalities associated with ACC are those of the posterior fossa, inter-hemispheric cysts and neuronal migrations disorders.

7. SHOULD MRI BE PART OF THE ASSESSMENT OF SUSPECTED ACC?

MRI allows direct visualization of the CC. This is important, as the diagnosis of ACC by ultrasound has been associated with a false positive rate ranging from 0-20%^{2,13,37,38}. In contrast, MRI can confirm ACC, assess whether this is complete or partial, and may also help in detecting coexisting brain abnormalities not seen on ultrasound, such as anomalies of gyration and heterotopia³⁹⁻⁴². In a recent systematic review, prenatal MRI was associated with detection of additional abnormalities in 22.5% of cases compared to USS¹.

The association of ACC with other cerebral anomalies increases the likelihood of a later neurological impairment, therefore MRI should be considered a part of ACC assessment, particularly in fetuses with apparently isolated ACC.

8. SHOULD PRENATAL KARYOTYPING BE OFFERED IN ACC?

ACC has been associated with several chromosomal rearrangements but it is difficult to be certain regarding the incidence of chromosomal abnormalities in ACC because studies are not consistent in reporting of chromosomal analysis⁴³. The overall rate of chromosomal abnormality was 17.8% (Appendix3), but it is important to note that this high rate includes both isolated and complex ACC. A recent study suggests that this high risk of chromosomal abnormalities is confined to complex cases⁴⁴. Chromosomal microdeletions may be associated with ACC and therefore chromosome analysis using Comparative Genomic Hybridisation (CGH) should be considered.

9. WHAT GENETIC SYNDROMES ARE ASSOCIATED WITH ACC?

ACC has been associated with several syndromes with autosomal dominant, autosomal recessive and X-linked mode of inheritance^{9,45}. An ONIM search for

“Agenesis of the corpus callosum” results in 238 entries. The more common genetic causes of callosal agenesis are:

Aicardi syndrome: this is almost exclusively found in girls as it is inherited as X-linked dominant trait. It is characterized by the presence of ACC associated with chorioretinal abnormalities, infantile seizures and mental retardation⁴⁶.

Andermann syndrome is a rare autosomal recessive disorder characterized by ACC, progressive motor-sensory neuropathy and mental retardation⁴⁷.

Given the uncertain prevalence of ACC makes it very difficult to establish the frequency of underlying genetic syndromes. Bedeschi et al studied 63 cases of ACC with a neuropsychiatric disorder (mental retardation of varying severity, learning disabilities or epilepsy) and found that 33% of the cases had a recognizable syndrome⁴⁸. Schell-Apacik studied 41 patients with ACC and found that 12% had a genetic syndrome⁹.

Apart from genetic syndromes agenesis of CC can be caused by non-syndromic conditions such as fetal alcohol syndrome and metabolic disorders⁴⁹.

10. SHOULD SCREENING FOR CONGENITAL INFECTION BE PERFORMED?

Although ACC in association with Cytomegalovirus, Toxoplasmosis, Rubella and Influenza virus has been reported in the literature, other associated abnormalities usually coexist; the value of screening for these infections in isolated ACC is uncertain⁵⁰⁻⁵³.

11. SHOULD SERIAL ANTENATAL EXAMINATIONS BE ARRANGED?

The prognosis in ACC is dependent on coexistence of other abnormalities; these may include abnormalities of cortical development, which can only be assessed with advancing gestation. Therefore additional detailed fetal ultrasound examination(s) should be considered in order to search for cerebral and extra-cerebral abnormalities that may not be evident during the second-trimester examination. There is no agreement in the literature regarding the timing and frequency of such follow-up in fetuses with ACC.

12. WHAT IS THE RISK THAT THE ACC IS NOT TRULY ISOLATED?

Structural abnormalities not detected prenatally may be apparent only on postnatal imaging or clinical examination. The ability of fetal imaging to correctly diagnose isolated ACC will vary depending upon a number of factors: the most important one is probably the time in gestation, as it is unlikely that some anomalies, such as cortical malformations, will be detected in the second trimester; other variables include: the clarity of ultrasound imaging achieved (which may be limited, for example, by maternal habitus) and expertise of the operators involved; the imaging protocol used; and use of MRI. On the other hand, the rigour, timing and length of post-natal follow-up will of course have a significant influence on the final diagnosis. Thus a precise estimate of the discrepancy between prenatal diagnosis and postnatal assessment of ACC from may be subject to some bias depending on the quality of the studies.

With all these limitations in mind, our review suggests that overall 15.1% of the cases thought to be isolated prenatally were found to have associated abnormalities after birth, highlighting the importance of postnatal assessment and follow up (Appendix 4).

11. WHAT IS THE NEUROLOGICAL OUTCOME OF CHILDREN WITH A PRENATAL DIAGNOSIS OF ISOLATED ACC?

A recent systematic review assessed the rate of neurodevelopmental outcome in 132 fetuses (16 studies) with isolated ACC. The authors reported neurodevelopmental outcome as normal; borderline or moderate disability; and severe disability. In complete ACC the respective figures were 74.3%, 14.3% and 11.4% while for isolated partial ACC they were 65.5%, 6.9% and 27.6%¹.

When taking into account only those studies providing the use of MRI and standardized tools of neurodevelopmental assessment, in complete ACC the rates were 83.7%, 8.2% and 8.2% for normal, borderline/moderate and severe disability, respectively. The corresponding rates for partial ACC were not reported in view of the small number of cases. The authors found no statistically significant difference in the outcome between fetuses with isolated complete and partial ACC¹.

The authors of the review highlighted many limitations in existing studies: limited and inconsistent data that prevented subgroup analyses, for example between partial and complete absence. Therefore providing a precise estimate of the risk of neurodevelopmental delay is difficult. In our view, one important limitation with available studies is the time of follow-up: in most cases, assessment was made in the pre-school period. This may represent an important shortcoming as in one series a progressive decline of intellect has been noted, with a considerable number of infants demonstrating learning difficulties in school⁵⁴⁻⁵⁵.

12. WHAT ARE THE LIMITATIONS OF THE AVAILABLE STUDIES?

- Study design: Most of the studies report a small number of cases collected retrospectively.
- Population selection: Most studies report on a referred population. However, cases of ACC with few indirect signs are less likely to be detected during screening and will not be referred. This means that studies are likely to include more severe cases, and this could influence the reported risk of poor neurodevelopmental outcome.
- Disease definition: Complete and partial; and complex and isolated cases are often considered together.
- Imaging protocol: Antenatal assessment of ACC is different between studies and not all of them include MRI evaluation.
- Ascertainment bias: There are high rates of pregnancy termination and this may influence outcome, with more severe cases likely to undergo termination. In addition many studies report high proportions of cases lost to follow-up.
- Outcome assessment and reporting: Use of subjective tests to assess the neurological development and lack of distinction in the severity of neurodevelopmental delay.
- Length of follow up: many studies report follow up soon after birth, and such assessment may not be sufficiently sensitive to detect abnormal neurological function.
- Lack of control groups to compare outcomes with those of the normal population.

CONCLUSION

Although the precise incidence is unknown, absence of the CC appears to be a rare condition. It may be suspected at the time of the routine anomaly scan due to the presence of indirect features, or diagnosed on direct visualisation. However, it is likely that in isolation it may remain undetected. The finding should trigger detailed assessment to establish whether the finding is isolated, or if there are associated ultrasound abnormalities. Careful evaluation of fetal anatomy should be carried out in an expert setting, and this should include multiplanar neurosonography, and transvaginal imaging may be required for this. Where available, fetal MRI is indicated, and this allows confirmation of the finding, and allows assessment of coexisting brain abnormalities. In part due to the rarity of the condition there remain many gaps in our knowledge: it is difficult to establish the probability for chromosomal abnormalities, but it appears that - in cases where there are no coexisting abnormalities - chromosomal abnormalities are uncommon. Similarly the risk of there being an underlying congenital infection appears low if there are no associated features. The rate of neurodevelopmental delay in infants with a prenatal diagnosis of isolated ACC is about 25-30%, and this appears to be similar in complete or partial agenesis. However, it must be recognised that there are many limitations in the studies that have reported on this condition, and such inconsistent data mean it is difficult to provide a precise estimate of the risk of neurodevelopmental delay. Given this, both antenatal and postnatal follow-up should be offered considered when this diagnosis is made.

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BOX 1: Some key points on indirect features of ACC**ABSENCE OF CAVUM SEPTI PELLUCIDI (CSP)**

- Not specific to ACC; also associated with holoprosencephaly, hydrocephalus, septo-optic dysplasia, schizencephaly, encephalocele, porencephaly and hydranencephaly^{32,33}(Fig. 2,4)
- In partial agenesis of the CC the CSP is usually present² (Fig.2, 4)

ABNORMALITIES OF THE VENTRICLES

- Colpocephaly (dilatation of the atria and occipital horns of the lateral ventricles on the standard axial view at the transventricular level); this is the result of the absence of the posterior portion of the CC, which allows expansion of the occipital horns. It is usually not associated with progressive ventriculomegaly^{13,34}(Fig. 4).
- Lateral displacement of the bodies of the lateral ventricles on coronal views due to a paired aberrant bundle of fibres (Bundle of Probst) that fail to cross the hemispheres and run parallel to the midline. These paired structures bulge the medial borders of the frontal horns that assume the shape of a bull's horn¹³ (Fig. 2, 4).
- Upward displacement of the third ventricle that reaches the level of the lateral ventricles (Fig.2, 4).

ABNORMAL COURSE OF PERICALLOSAL ARTERY

- Complete ACC: the semicircular loop of the pericallosal artery is lost and the branches of the anterior cerebral artery ascend linearly¹³.
- Partial ACC: the pericallosal artery follows the anterior part of the CC but then loses its normal course where the CC disappears posteriorly; at this level the artery takes an upward posterior oblique direction^{2,31}.

WIDENING OF THE INTER-HEMISPHERIC FISSURE

- The absence of CC frequently results in increased separation of the hemispheres with a prominent inter-hemispheric fissure. In this condition USS will reveal three echogenic lines running parallel in the upper cranium: the middle one representing the falxcerebri and the lateral ones representing the medial borders of the separated hemispheres¹³(Fig.2,4)

RADIAL DISPOSITION OF THE SULCI ON THE INTERNAL ASPECTS OF THE HEMISPHERES

- The absence of the CC results in an abnormal induction of the medial cerebral convolutions determining a radiate arrangement of cerebral sulci around the roof of the third ventricle¹³(Fig.3).

Figures:

Fig.1: Normal appearance of the corpus callosum (CC) in a median view of the brain: a) anatomic specimen from an adult brain; b,c) fetal sonograms; d) fetal magnetic resonance. Labels are 1) rostrum; 2) genu; 3) body; 4) splenium; 5) cavumseptipellucidi (CSP); 6) cavumVergae (CV); 7) pericallosal artery.

Fig. 2: Sonography of a normal brain, complete and partial callosal agenesis in axial (a-c), coronal (d-f) and median (g,i) views of the fetal head in the second trimester of gestation. With complete agenesis, the anatomical complex formed by CC and CSP is completely absent, the inter-hemispheric fissure is enlarged, and the frontal horns (FH) are more separated than normal. With partial agenesis, the findings are more subtle, and the CC and CSP are present but shortened (3v = third ventricle).

Fig. 3: Magnetic resonance imaging of agenesis of the corpus callosum in a third trimester fetus in (a) axial, (b) median and (c) coronal views. Apart from demonstrating the absence of the corpus callosum, magnetic resonance has the advantage of clearly showing the normality of the convolutional pattern, a finding that has a major impact on prognosis. Notice the radial arrangement of the gyri around the roof of the third ventricle (3v).

Fig. 4: A summary of sonographic findings that are associated with agenesis of the corpus callosum in the axial plane: 1) the CSP is not demonstrated and its position is occupied by three lines that are due to the widened inter-hemispheric fissure, centrally partitioned by the falxcerebrii; 2) the frontal horns of lateral ventricles are further away from the midline (laterally displaced) and the atria of lateral ventricles slightly enlarged; this results in a 'tear-shaped' appearance of the lateral ventricles; 3) an inter-hemispheric cyst is seen; 4) an echogenic structure is seen in the position normally occupied by the CSP; this is a lipoma, that is frequently associated with partial or complete callosal agenesis but is usually visible only in late gestation.

APPENDIX 1

Search strategy used for the literature review in this article:

A search was conducted of an electronic database (Pubmed) on 5/5/2011 and updated on 31/07/2012

- “corpus callosum” “agenesis” = 3545
- “corpus callosum” “hypoplasia” = 3393
- “corpus callosum” “dysgenesis” = 2120
- “corpus callosum” “hypogenesis” = 48
- “corpus callosum” “anomalies” = 3414
- “corpus callosum” “anomaly” = 214
- “corpus callosum” “fetus” = 221
- “corpus callosum” “fetal” = 656
- “corpus callosum” “prenatal” = 490
- “corpus callosum” “neurodevelopment” = 30
- “corpus callosum” “neurodevelopmental” = 129

A combination of the 4 searches yielded 4278 articles; the search was then limited to English language yielding 3668 articles. Articles before 1980 were excluded for outcome evaluation as we assumed that antenatal imaging techniques would not reflect current practice. For the purpose of each analysis series with less than three cases were excluded.

APPENDIX 2: Papers reporting on the incidence of coexisting brain abnormalities in ACC diagnosed antenatally.

NOTE TO EDITOR: THIS COULD APPEAR AS ONLINE CONTENT

Author	Year	Cases, n	Cases with other brain abnormality, n (%)	ACC	Imaging
Li ^{44b}	2012	41	27 (66)	Complete	US/MRI
Ghi ³¹	2010	14	5 (36)	Partial	US/MRI/Autopsy
Tang ⁵⁶	2009	29	27 (93)	Complete	MRI/Autopsy
Fratelli ³⁸	2007	117	30 (26)	Complete	US/MRI
Volpe ²	2006	19	8 (42)	Partial	US/MRI/ Autopsy
Glenn ³⁷	2005	8	8 (100)	Both	US/MRI/ Autopsy
Sonigo ⁴⁰	1998	50	30(60)	Both ^d	US/MRI
D'Ercole ³⁹	1998	14	9 (64)	Both	US/MRI/ Autopsy
Bennett ^{36c}	1996	14	3 (21)	Both	US/CT/MRI
Vergani ¹¹	1994	14	5 (36)	Both	US/MRI/Autopsy
Pilu ^{13a}	1993	35	10 (29)	Both	US
Blum ⁵⁸	1990	16	8(50)	Both	US/TC/MRI/ Autopsy
Bertino ⁵⁹	1988	7	3 (43)	Both	US/TC/Autopsy
Total		378	173(45.8)		

a. Includes the paper by Sandriet al.⁵⁷

b. Only includes cases of complete ACC

c. Excludes one case of holoprosencephaly

d. Not clearly specified whether ACC or other callosal abnormalities

APPENDIX 3: Papers reporting the incidence of chromosomal abnormalities in fetuses with ACC.

NOTE TO EDITOR: THIS COULD APPEAR AS TABLE IN THE MANUSCRIPT

Author	Year	Cases (n) ^a	Type ACC	Chromosomal abnormalities n (%)	Trisomy 8 n (%)	Trisomy 13 n (%)	Trisomy 18 n (%)	Trisomy 21 n (%)	Triploidy n (%)	Others n (%)
Li ⁴⁴	2012	41	Complete	12 (29.3)	3 (25)	-	1 (8.3)	-	-	8 (66.7)
Cignini ⁶⁰	2010	17	Complete	1 (5.9)	1(100)	-	-	-	-	-
Chadie ⁵³	2008	13	Both	0 (0)	-	-	-	-	-	-
Schell-Apacik ⁹	2008	41	Both	8 (19.5)	2 (25)	-	-	-	-	6 (75)
Fratelli ³⁸	2007	117	Complete	33 (28.2)	-	1 (3)	23 (69.7)	1 (3)	2 (6)	6 (18.1)
Bedeschi ⁴⁸	2006	62	Both	7 (11.3)	-	-	-	1 (14.3)	-	6 (85.7)
Volpe ²	2006	18	Partial	3 (16.7)	-	-	2 (66.7)	-	-	1 (33.3)
Pisani ⁵¹	2006	9	Complete	0 (0)	-	-	-	-	-	-
Dos Santos ⁶¹	2002	17	Complete	3 (17.6)	3 (100)	-	-	-	-	-
Shevell ²⁴	2002	24	Both	3 (12.5)	-	-	-	-	-	3 (100)
Marszal ⁶²	2000	7	ns	2 (28.6)	-	1 (50)	-	-	-	1 (50)
D'Ercole ³⁹	1998	14	Both	0 (0)	-	-	-	-	-	-
Benett ³⁶	1996	10	Both	2 (20)	-	-	1 (50)	-	-	1 (50)
Vergani ¹¹	1994	14	Both	1 (7.1)	-	1 (100)	-	-	-	-
Pilu ^{13a}	1993	29	Both	6 (20.7)	3 (50)	-	2 (33.3)	-	-	1 (17.7)
Serur ⁴³	1988	33	Both	2 (6.1)	2 (100)	-	-	-	-	-
Total		466		83 (17.8)	14 (3)	3 (0.6)	29 (6.2)	2 (0.4)	2 (0.4)	33 (7.1)

a. Includes the paper by Sandriet al.⁵⁷

APPENDIX 4: Papers reporting on the rate of associated abnormalities detected postnatally in fetuses diagnosed prenatally as having isolated ACC (m= months; y= years)

NOTE TO EDITOR: THIS COULD APPEAR AS TABLE IN THE MANUSCRIPT

Author	Year	Cases ^a	Type of prenatal examination	Duration of follow-up (range)	Diagnosed as not isolated postnatally
Li ⁴⁴	2012	9	USS and MRI	1-5 y	3 (33.3)
Moutard ^{54,55b}	2003/2012	17	USS and MRI	3-10y	0(0)
Mangione ⁶³	2011	23	USS and MRI	2,5-6,2 y	4 (17.4)
Ghi ³¹	2011	4	USS and MRI ^c	1-10 y	1 (25)
Manfredi ⁶⁴	2010	6	MRI	-	1 (16.6)
Cignini ⁶⁰	2010	13	USS	4 y	0 (0)
Pisani ⁵¹	2009	7	USS	2-16 y	1(16.6)
Chadie ⁵³	2008	17	USS and MRI ^c	3-16 y	4 (23.5)
Fratelli ³⁸	2007	11	USS and MRI ^c	1-5 y	0 (0)
Ramelli ⁶⁵	2006	3	USS	2-10 y	0 (0)
Volpe ²	2006	9	USS and MRI ^c	1-6 y	1 (11.1)
Bennett ³⁶	1996	10	USS	6 m -6 y	3 (30)
Vergani ¹¹	1994	7	USS	6 m-8 y	2 (28.6)
Pilu ^{13d}	1993	11	USS	6 m-11 y	0 (0)
Bertino ⁵⁹	1988	5	USS	0m-5 y	3 (60)
Total		152		0 m-16y	23(15.1)

a: Included only live births, neonatal deaths < 1month were excluded.

b: Same study population with different periods of follow-up.

c:Prenatal MRI not available for all the study population.

d: Includes the paper by Sandri et al.⁵⁷







